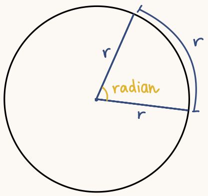
PURE 4.

Circular measure



- The arc of the angle 1 radian = length of radius

- circumference = $2\pi r$

  ↳ $360^\circ = 2\pi$ radians

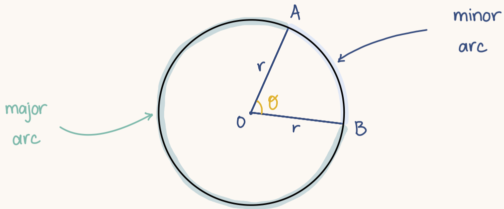
Degrees	0°	30°	45°	60°	90°	180°	270°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π

* conversion:

- $a^\circ \longrightarrow$ rad : $a \times \frac{\pi}{180^\circ}$
- b rad $\longrightarrow$ ° : $b \times \frac{180^\circ}{\pi}$

(* 1 rad = $1 \times \frac{180^\circ}{\pi} \approx 57^\circ$)

A) Arc length

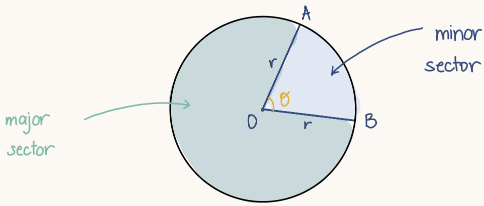


- when θ measured in radians

  ↳ length of AB = $r\theta$

(arc length = $\frac{\theta}{\cancel{2\pi}} \times \cancel{2\pi} r = r\theta$)

B) Area of a sector



- when θ measured in radians

  ↳ area of sector AOB = $\frac{1}{2} r\theta^2$

(area = $\frac{\theta}{2\cancel{\pi}} \times \cancel{\pi} r^2 = \frac{1}{2} r^2 \theta$)